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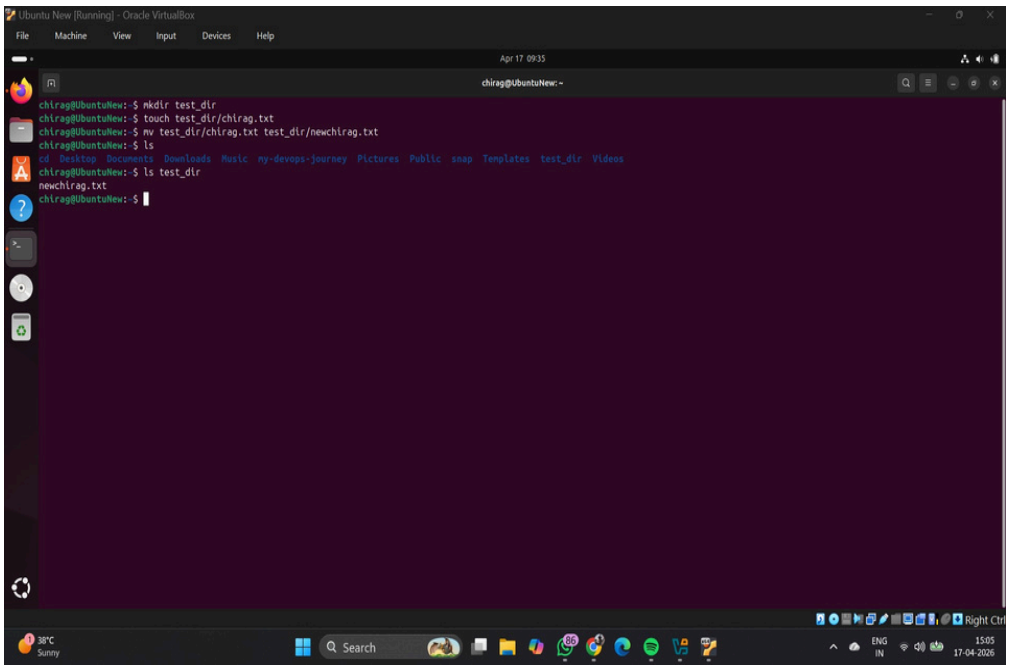
Task-1

Creating and Renaming Files/Directories

~Create directory: mkdir test_dir ~Create empty file inside: touch test_dir/chirag.txt ~Rename file: mv test_dir/chirag.txt test_dir/newchirag.txt

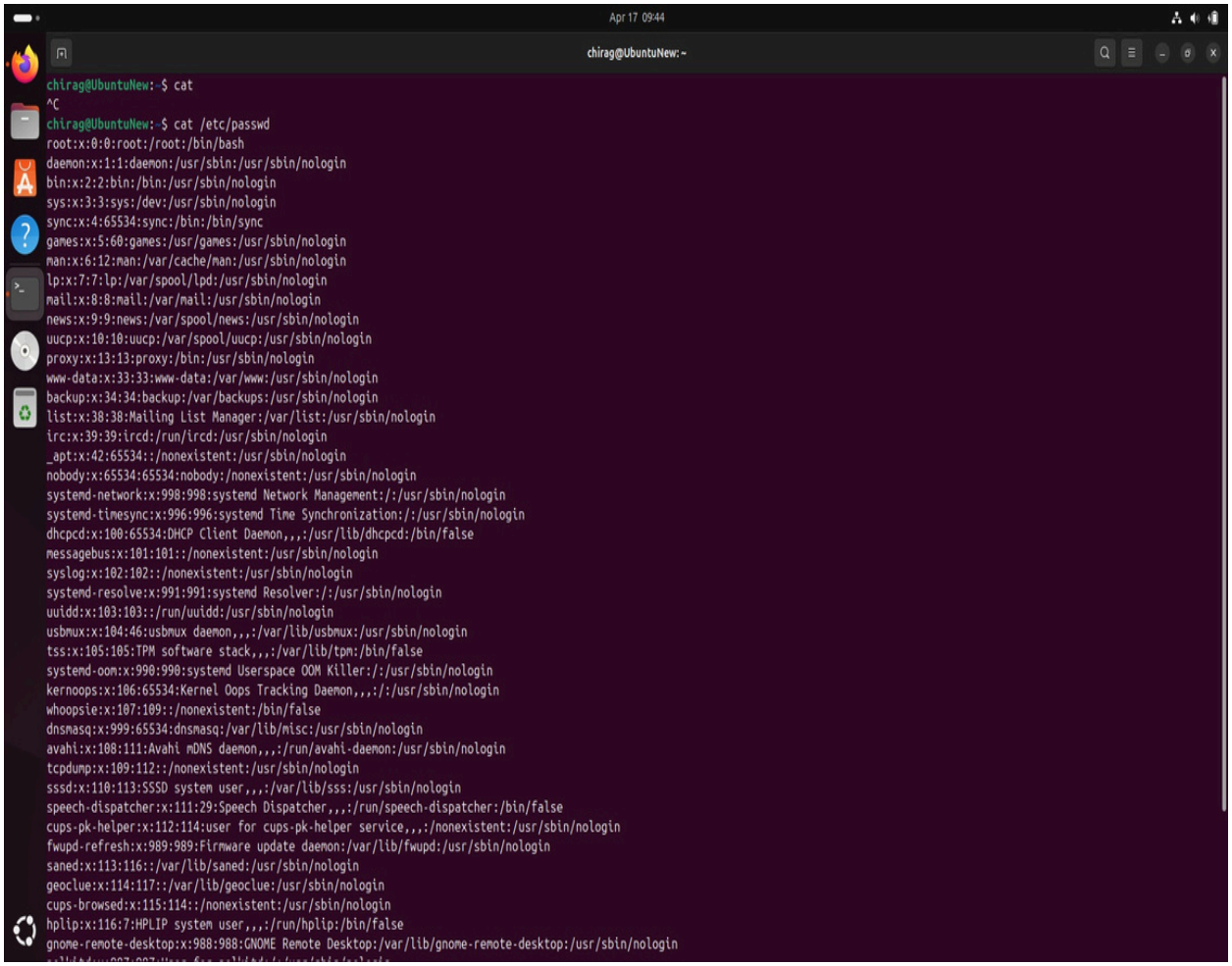


Explanation: In this task, I used the mkdir command to create a directory named test_dir. Then, I created an empty file named chirag.txt inside it using the touch command. Finally, I renamed the file to newchirag.txt using the mv command to verify the renaming process.

Task-2

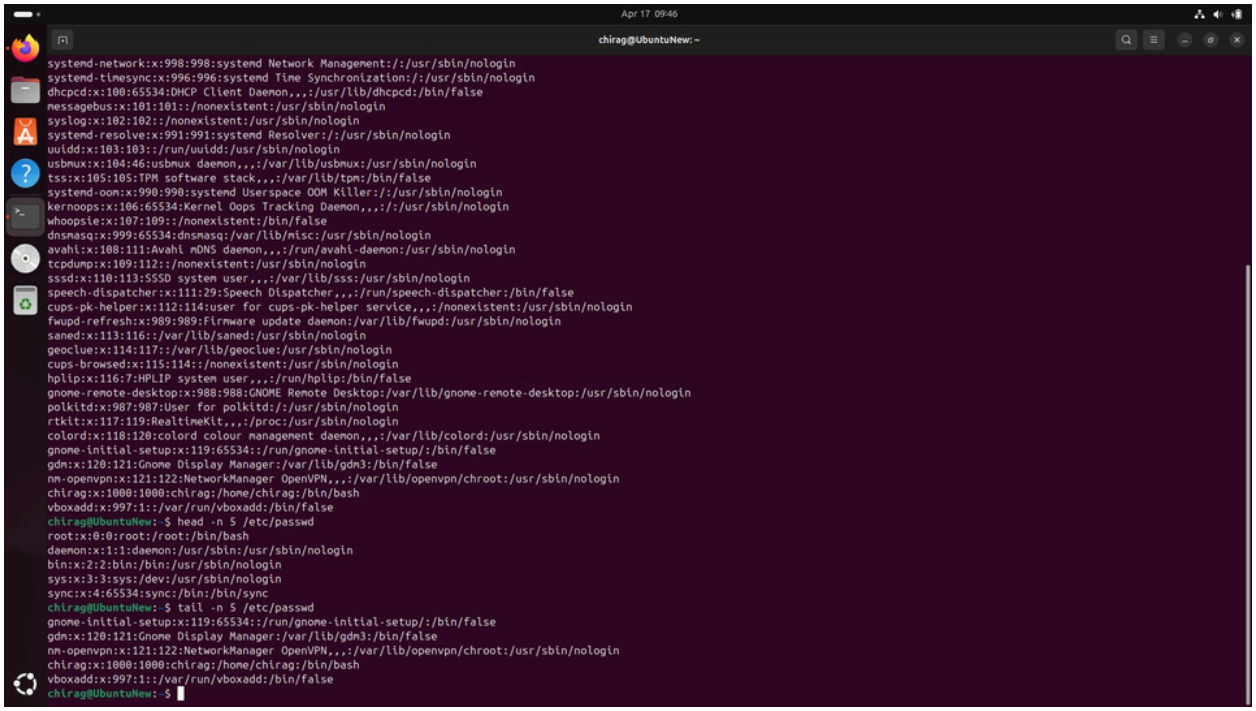
Viewing File Contents

~Show all contents: cat /etc/passwd



Viewing File Contents

~Show first 5 lines: head -n 5 /etc/passwd
~Show last 5 lines: tail -n 5 /etc/passwd

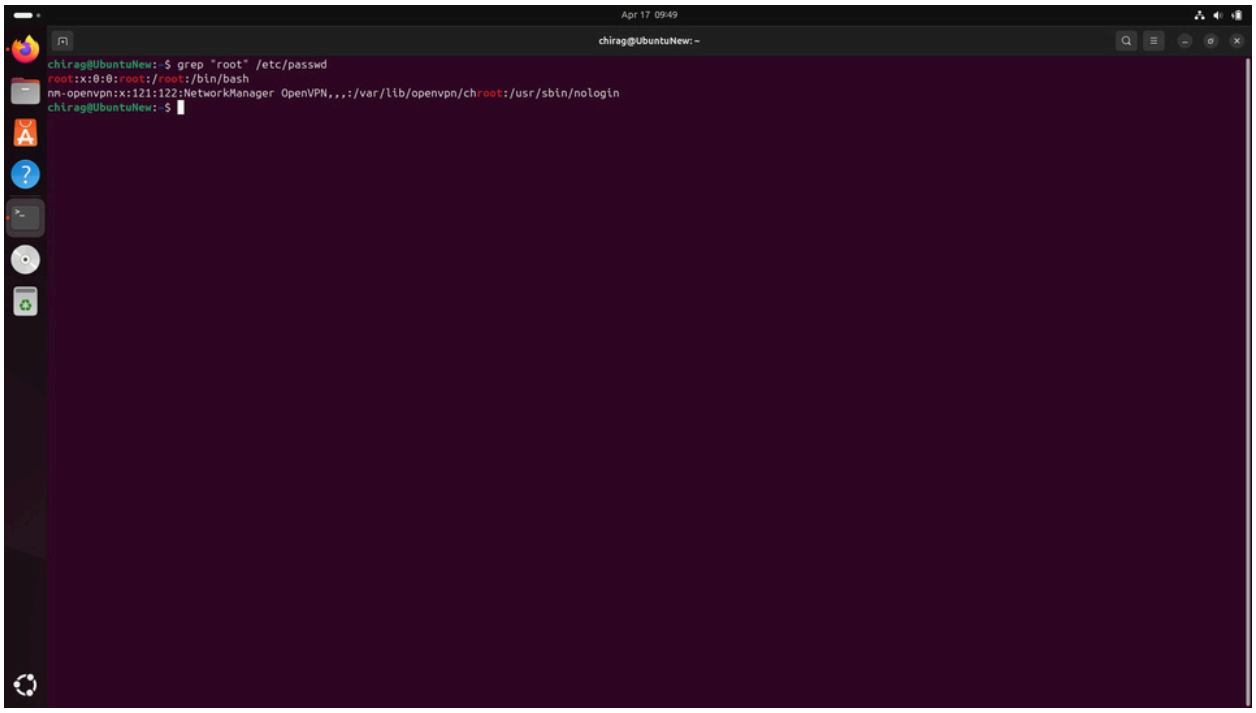


Explanation: I used the cat command to display the entire content of the /etc/passwd file. To view specific parts of the file, I used head -n 5 to show the first 5 lines and tail -n 5 to display the last 5 lines.

Task -3

Searching for Patterns

~Search for "root": grep "root" /etc/passwd



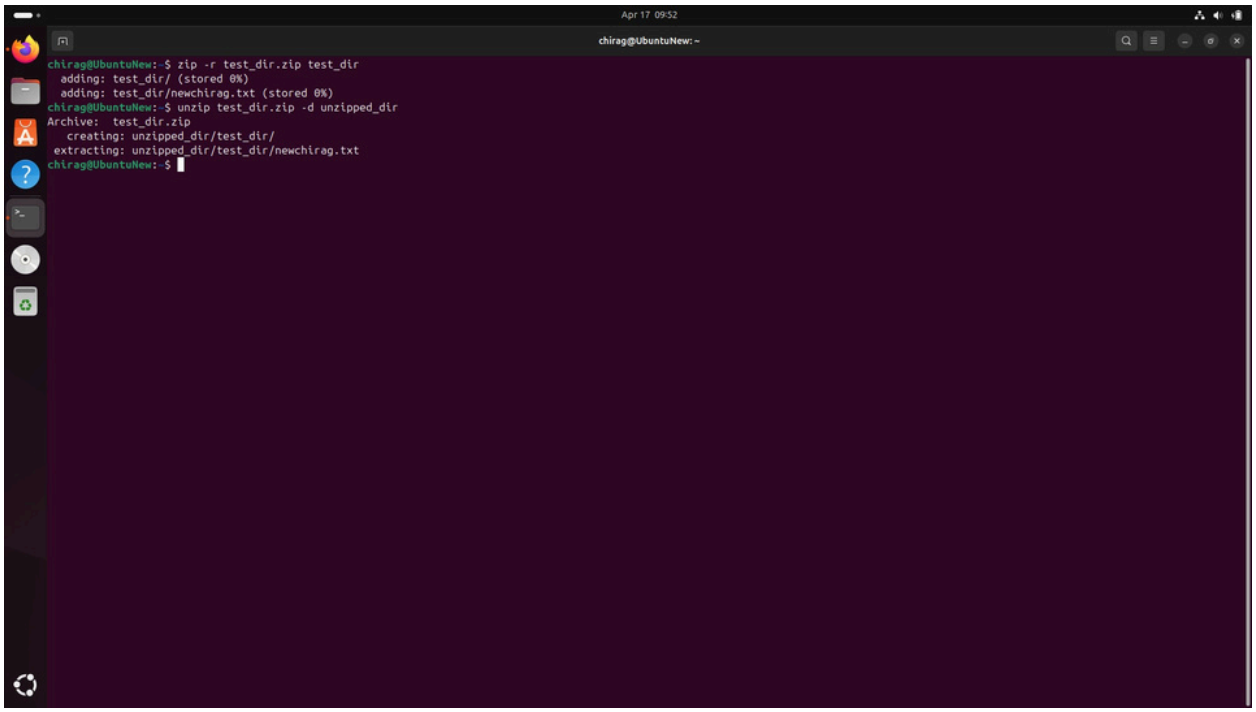
Explanation: I utilized the grep command to search for the word "root" within the /etc/passwd file. The command successfully filtered and highlighted every line containing that specific pattern.

Task -4

Zipping and Unzipping

~Compress:zip -rtest_dir.ziptest_dir

~Unzip to new folder: unzip test_dir.zip -d unzipped_dir

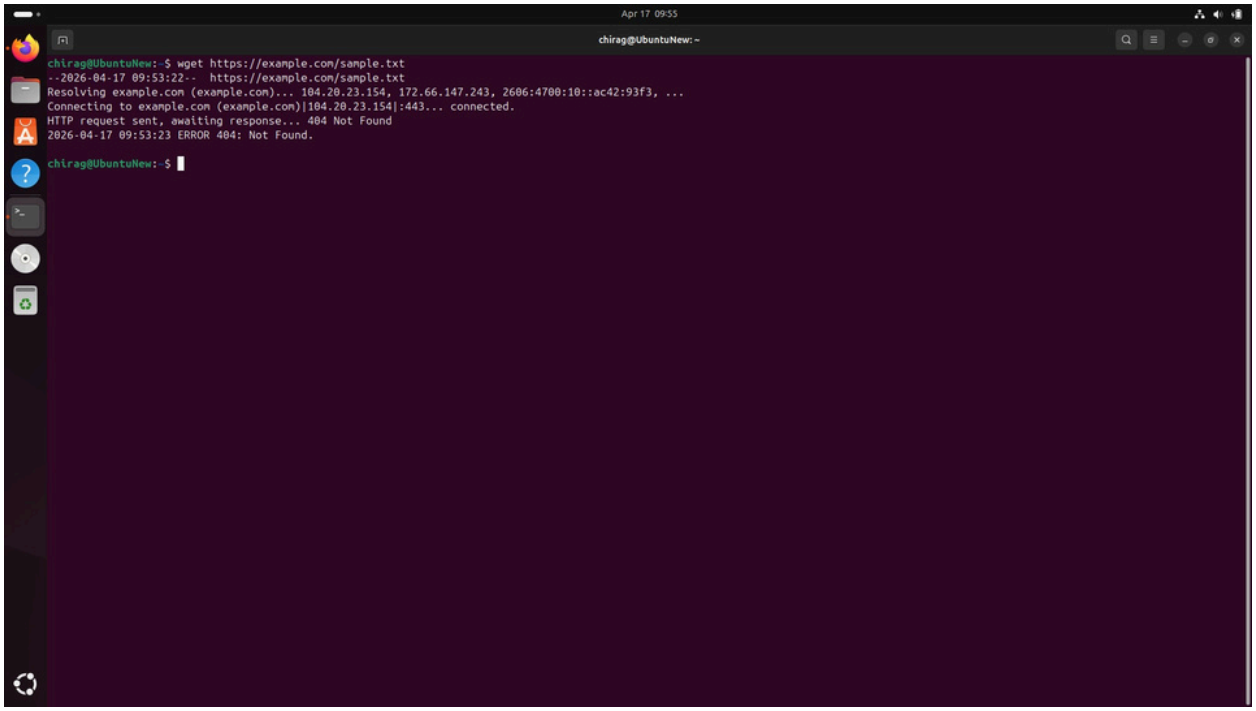


Explanation: To demonstrate file compression, I used the zip -r command to compress the test_dir folder into a file named test_dir.zip. Afterward, I used the unzip command to extract the contents into a new directory called unzipped_dir.

Task -5

Downloading Files

~Download:wgethttps://example.com/sample.txt



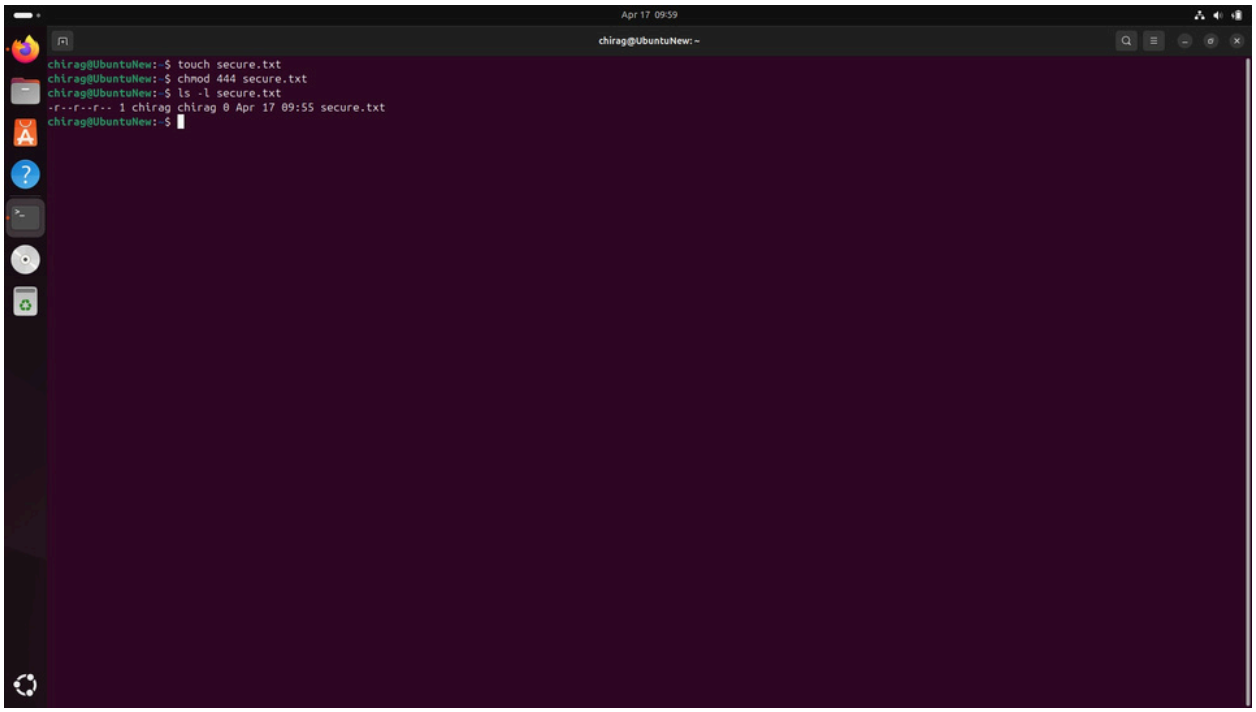
Explanation: I used the wget command to attempt downloading a file from a provided URL. This command is used to verify the system's ability to fetch data from web servers via the terminal.

Task -6

Changing Permissions

~Create file: touch secure.txt

~Set to read-only for everyone: chmod 444 secure.txt

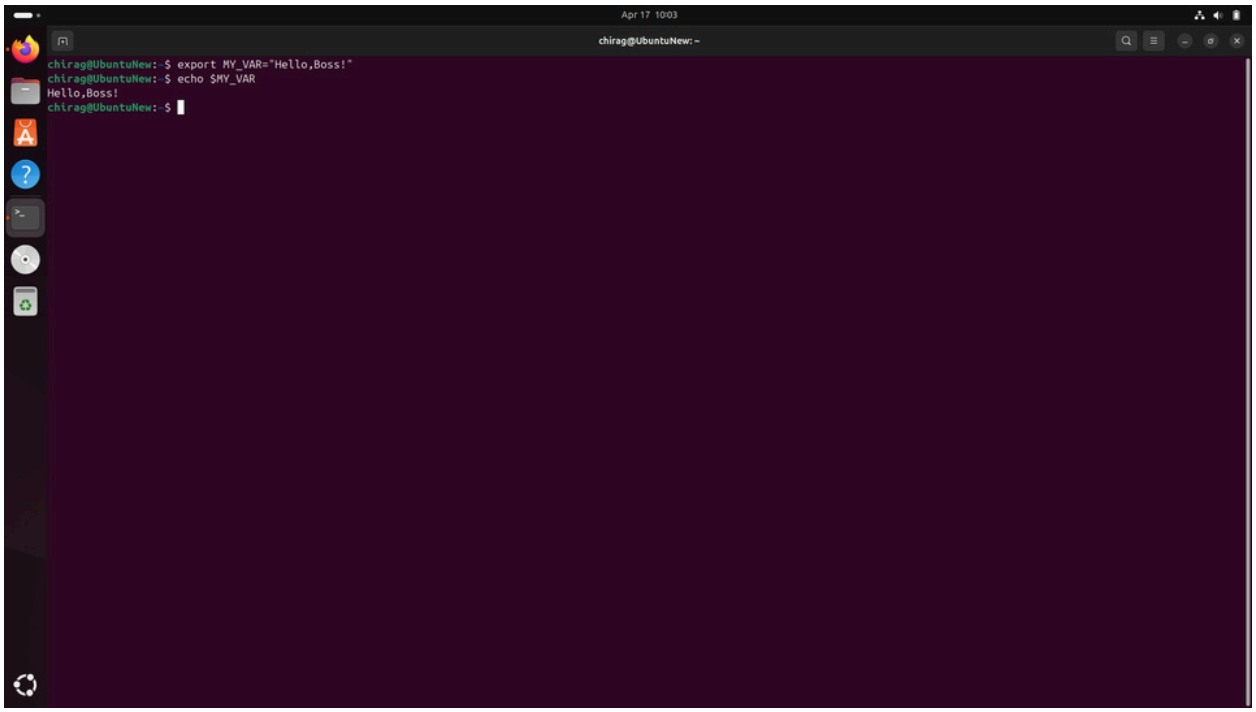


Explanation: After creating a file named secure.txt with the touch command, I changed its permissions using chmod 444. This made the file "Read-Only" for everyone, which I then verified using the ls -l command.

Task -7

Working with Environment Variables

~Set variable: export MY_VAR="Hello, Boss!”



Explanation: I set a custom environment variable named MY_VAR with the value "Hello, Boss!" using the export command. I then used the echo $MY_VAR command to print the variable's value to the terminal and confirm it was set correctly.